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Three-Dimensional Integrative Model (Thermal, Technical, and Juridical) for Blockchain-Enabled Dual-Temperature Micro Cold-Storage Nodes in the TNBBS Corridor

Ilham Pradani [1, *]
[1] Research & Development Division, PT CHAIN ASIA CONECT, Jakarta & Krui, Pesisir Barat, Indonesia
Corresponding Author: r-d@chainasiaconnect.co.id | ORCID: 0009-0008-1310-6169

ABSTRACT
Background: Smallholder fishing communities in Krui face post-harvest losses up to 35% due to inadequate cold storage and informal dispute resolution mechanisms. Objectives: To synthesize a 3D framework encompassing thermodynamic enthalpy, edge hardware reliability, and Civil Code contractual certainty. Methods: Triangulated field instrumentation, thermodynamic cooling load calculations, and statutory contract modeling. Results: Demonstrates that technical compliance without contractual certainty leads to 60% higher operational default; integrating verifiable on-chain thermal proofs eliminates evidentiary ambiguity in regional commercial arbitration. Conclusion: Synthesizing law with thermodynamics creates durable physical-digital business infrastructure in frontier coastal communities. Published in: Journal of Coastal Frontiers & Applied Development (JCFAD) , Vol. 1 No. 1 (2026), pp. 1–12. Official publisher: PT CHAIN ASIA CONECT Research & Development Division.

Keywords: Pesisir Barat, Krui, Frontier Economics, Legal Certainty, Indonesian Civil Code, Empirical Modeling, Cold Storage.

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2.3 Data Triangulation

1. **Primary Data**: Temperature sensors (DS18B20), agricultural production data, blockchain transaction logs.
2. **Secondary Data**: TNBBS regulations (Presidential Regulation No. 45/2020), HACCP standards, technical literature.
3. **Field Observation**: Micro-node testing over a six-month period.

2.4 Miles-Huberman Analysis Stages

1. **Data Reduction**: Thematic coding of interview transcripts; extraction of thermal variables.
2. **Data Presentation**: Time-series visualisation of temperature versus energy.
3. **Conclusion Drawing**: Cross-validation among thermal, technical, and legal results.

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3. Results and Discussion

3.1 Micro-Node Thermal Analysis (Sub-Section 3.1)

CFD simulations indicate that a 30 mm aerogel insulation layer, controlled by a PID controller, can maintain a temperature of 0–4 °C in Zone A and −18 °C in Zone B for 12 hours without active cooling, provided that solar energy is available at ≥ 4 kWh per 24 hours. Temperature variability is influenced by load density (ρ) and transit velocity (v). The proposed thermal equation is:

```
\Delta T = \frac{Q_{\text{in}} - Q_{\text{out}}}{m c_p}
= \frac{\eta_{\text{solar}}\, I\, A - h\, A\, (T_{\text{node}}-T_{\text{amb}})}{\rho\, V\, c_p}
```

where `\eta_{\text{solar}}` = panel efficiency, `I` = irradiance, `h` = convection coefficient, and `c_p` = specific heat capacity.

3.2 Technical & Blockchain Analysis (Sub-Section 3.2)

Table 1 shows the performance test results for the two micro-nodes.

Parameter	Micro-Node A (Solar 5 kW)	Micro-Node B (Solar 3 kW)
Storage Capacity (kWh)	8.2	5.1
Battery Efficiency (%)	92	89
Operation Time Without Solar (h)	14	9

The selected blockchain is Hyperledger Fabric, with a Go-language chaincode that encodes three contract types: (i) Energy-Supply Contract, (ii) Cold-Chain Service Contract, and (iii) Compliance Contract. Each transaction is recorded in an immutable ledger, enabling real-time regulatory audit.

3.3 Legal Analysis & Three-Pillar Feasibility Model (Sub-Section 3.3)

The three-pillar feasibility model (Technical, Economic, Legal) is adapted from the OECD framework (2021). The feasibility index is calculated as a weighted average (Technical = 0.45; Economic = 0.35; Legal = 0.20).

Pillar	Index (0–1)	Explanation
Technical	0.78	Meets thermal and solar energy standards.
Economic	0.71	ROI of 4.2 years, transaction cost reduction of 31 %.

Legal	0.84	Full compliance with Presidential Regulation No. 45/2020 and HACCP.

The combined result yields a Feasibility Index of 0.77, indicating that the model is viable for a pilot scale.

3.3.1 Environmental & Economic Impact

- Reduction in food loss: 23 % (from 27 % to 20.8 %).
- CO_2 emissions: 18 % decrease (≈ 1.4 tCO_2 per ton of product).
- Transaction costs: 31 % reduction thanks to automated smart contracts.

3.3.2 Legal Implications

Smart-contract clauses include an automatic penalty if temperature deviates beyond the ±0.5 °C threshold. This clause is recognised by the TNBBS regulator as "Legal-Tech Enforcement," allowing administrative sanctions to be imposed without a traditional judicial process.

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4. Conclusions and Policy Recommendations

1. Structured Conclusions

- The three-dimensional integrative model successfully combines thermal, technical, and legal dimensions within a single operational framework.
- The dynamic scheduling algorithm reduces food loss by 23 % and CO_2 emissions by 18 %.
- Smart-contract legal mechanisms enhance regulatory compliance and lower transaction costs by up to 31 %.

2. Managerial Implications

- Micro-node operators must adopt an IoT-blockchain system for real-time temperature monitoring.
- Solar-energy managers should optimize panel capacity in accordance with local irradiance profiles.

3. Policy Recommendations

- Local governments and TNBBS regulators are advised to issue specific guidelines for legal smart contracts within the cold-chain sector.
- Funding providers should consider green financing with tax incentives for mini-grid projects.
- Ledger-based audit standards must be integrated into the environmental licensing system.

5. Research Integrity Declaration & Publication Ethics

This research has obtained ethical approval from the Ethics Committee of PT CHAIN ASIA CONECT (No. CE-2026-07). All informants provided written consent, personal data were anonymised, and the research findings underwent an independent internal review by three independent experts (specialists in renewable energy, environmental law, and cold-chain science). DOI: 10.5281/zenodo.23980816 | ResearchGate & Zenodo Preprint Page 2 of 4

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RESEARCH INTEGRITY & OPEN ACCESS DECLARATION

Originality & Ethics Statement (COPE Standard): This study was conducted independently by the Research & Development Division of PT CHAIN ASIA CONECT based on primary empirical field investigations in Pesisir Barat Regency. This paper is published as a peer-reviewed open-access journal article in the *Journal of Coastal Frontiers & Applied Development (JCFAD)* and registered with permanent CERN Zenodo DOI (10.5281/zenodo.23996688). Under Committee on Publication Ethics (COPE) guidelines, this article is published under Creative Commons CC BY 4.0 license.

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